

Supplementary Evidence for Fed-MDBSCAN-G: Audit-v2 Revision

Working companion to the main manuscript

TABLE VI
CANONICAL FAILED UNITS. HAR: SAMPLE-WEIGHTED MEAN; CIFAR:
HDBSCAN FLAME.

Dataset	Condition	Seed
HAR	3.1	42
HAR	3.1	2024
HAR	3.2	137
HAR	3.2	2024
CIFAR-10	7.1	42

I. SCOPE AND PROVENANCE

This supplement describes the canonical audit-v2 matrix and diagnostic repetitions separately. Tables continue the numbering of the main manuscript. Numerical data are copied without changing the canonical run files; SHA-256 hashes are listed in `evidence/provenance.json`. The scenario manifest enumerates datasets, methods, seeds, attack parameters, and overrides. A reported standard deviation in the CSVs is over seeds, not independent rounds.

The canonical plan contains 2,130 units, 2,125 valid and five failed. A unit denotes a method, condition, and seed. Missing successful JSON output is not evidence that an experiment was never attempted: outcomes distinguish failure from missing work. Table VI lists the persistent failed units. Isolated numerical failures are retained, with no zero-accuracy imputation or seed replacement.

II. TEMPORAL ALARM BEHAVIOR

The temporal block activates the attack in zero-based rounds 10–19. Ten preceding and ten following rounds are clean. The table reports final accuracy and post-attack alarm FPR, averaged over three seeds. Optimizer momentum remains unchanged when alarm memory is removed. Persistent alarms after attack removal are false positives, not successful detection of a continuing attack. Identical final accuracies do not imply identical alert behavior.

III. EQUAL LOCAL-STEP CONTROL

The fixed-step block limits local optimization to five steps. This changes the training protocol and must not be pooled with the epoch-based main matrix. The three compared methods share the same condition and seed within the block. Results below show that equalizing local steps does not uniformly eliminate the utility gap relative to uniform averaging.

TABLE VII
TEMPORAL BLOCK: ACCURACY AND POST-ATTACK ALARM FPR
(PERCENT).

Data/condition	Variant	Acc.	Post FP
MNIST 2.1	Full	94.07	30.0
MNIST 2.1	No memory	94.07	0.0
MNIST 3.1	Full	48.25	100.0
MNIST 3.1	No memory	48.25	100.0
Fashion 2.1	Full	83.38	30.0
Fashion 2.1	No memory	83.38	0.0
Fashion 3.1	Full	35.44	100.0
Fashion 3.1	No memory	35.44	100.0
HAR 2.1	Full	56.43	100.0
HAR 2.1	No memory	56.81	93.3
HAR 3.1	Full	41.32	100.0
HAR 3.1	No memory	41.32	96.7

TABLE VIII
FIVE-STEP FINAL ACCURACY (PERCENT), THREE SEEDS PER CELL.

Data/condition	Full	L0	Uniform
MNIST 3.3	52.22	52.22	52.22
MNIST 5.2	28.44	33.19	72.27
Fashion 3.3	52.33	52.33	52.33
Fashion 5.2	47.58	47.63	64.74
HAR 3.3	27.17	27.17	27.17
HAR 5.2	44.27	44.27	44.35

IV. HONEST-CLIENT SIZE GROUPS

Client-group false rejection is pooled over observed honest submissions, with the denominator printed explicitly. The size quartile is a recorded partition descriptor, not a demographic category. Counts reuse the same clients across rounds and are not independent observations. They characterize retention disparities; they do not establish causal fairness or minority learning benefit. Dominant-label breakdowns are available in the accompanying CSV.

V. BACKEND REPRODUCIBILITY STUDY

Normal-profile short repetitions share initial-model and partition hashes but differ at the first client update; 268 of 273 trace entries differ between the two normal repeats. Under deterministic execution, two short repeats and an instrumented short repeat have identical traces. Two further independent 30-round deterministic-profile runs match at all 2,730 trace entries and each ends at accuracy 0.0994.

These runs concern only CIFAR-10, condition 7.1, HDBSCAN FLAME, seed 42. The deterministic profile sets `CUBLAS_WORKSPACE_CONFIG=:4096:8`, enables deterministic PyTorch algorithms, enables cuDNN determinism, and disables cuDNN benchmarking. This is an execution-profile intervention, not a change promoted into the canonical

TABLE IX
FULL METHOD, ATTACK-FREE CONDITION 6.2: SIZE-QUARTILE HONEST
REJECTION. Q0 IS THE RECORDED LOWEST QUARTILE.

Group	FP	FP+TN	FPR (%)
MNIST Q0	90	2002	4.50
MNIST Q1	55	2047	2.69
MNIST Q2	1284	2027	63.34
MNIST Q3	2019	2024	99.75
Fashion Q0	45	2006	2.24
Fashion Q1	93	2051	4.53
Fashion Q2	1055	2015	52.36
Fashion Q3	1941	2028	95.71
HAR Q0	10	2037	0.49
HAR Q1	197	2019	9.76
HAR Q2	287	2027	14.16
HAR Q3	1480	2017	73.38
CIFAR Q0	100	2007	4.98
CIFAR Q1	77	2044	3.77
CIFAR Q2	1033	2024	51.04
CIFAR Q3	2025	2025	100.00

audit-v2 matrix. Trace equality does not prove cross-device repeatability or effective defense. The raw trace and comparison files remain in the investigation directory identified by the project report.

VI. EVIDENCE FILE GUIDE

- `units.csv`: successful-unit final metrics and identity fields.
- `outcomes.csv`: all planned unit outcomes, including failures.
- `cells.csv`, `cell_status.csv`: cell summaries and coverage; incomplete cells must retain their success counts.
- `paired_deltas.csv`, `paired_summary.csv`: identity-matched method/control contrasts; contrast direction is explicit.
- `benign_groups.csv`: honest-client rejection counts by group.
- `scenario_manifest.json`: exact 149-job plan.
- `provenance.json`: source identity and copied CSV hashes.

The preserve-empty and oracle blocks remain in these files. They are protocol sensitivity and evaluator-reference experiments, respectively; neither is silently pooled into the main method ranking. Full statistical inference, original-author baseline reproduction, public artifact deposition, and a complete novelty assessment remain author-review tasks before submission.